# Classical Field Theory — Exercise Sheet 5

*Cones with forcing and interior tails*

**Solutions**

Write $\Box = \partial_t^2 - \Delta$. Every cone argument below concerns real $C^2$ fields on an open neighbourhood of the closed cone $\{0 \le t \le R,\ |x-a| \le R-t\}$, with $R > 0$. No existence theorem for a nonlinear equation is assumed or proved here.

**Exercise 5.1 (Source work on a shrinking cone).** Suppose $(\Box + m^2)w = F$, with $m \ge 0$ and continuous $F$. Let $E_m(t)$ be the integral over $B_{R-t}(a)$ of $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$.

(a) Derive the exact shrinking-ball balance, including source work and all lateral flux terms. Prove, for $0 \le t < R$,

$$E_m(t) \le e^t \left[ E_m(0) + \frac{1}{2} \int_0^t e^{-s} \|F(s)\|^2_{L^2(B_{R-s}(a))} \mathrm{d}s \right].$$

(b) If $F = 0$ on the cone and both initial data vanish on its base, prove that $w = 0$ on the closed cone, including when $m = 0$. Explain why this is a local statement even if total energy is infinite.

**Solution.** The energy density and spatial flux are $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$ and $f = -w_t \nabla w$. Differentiating and using the forced equation gives

$$\partial_t e_m + \operatorname{div} f = w_t (w_{tt} - \Delta w + m^2 w) = w_t F.$$

Lecture 5's moving-ball and ball-divergence formulas apply to these $C^1$ quantities. With $n$ the outward spatial normal they give

$$E_m'(t) = \int_{B_{R-t}(a)} w_t F \, \mathrm{d}^d x - \int_{\partial B_{R-t}(a)} (e_m + f \cdot n) \mathrm{d}S.$$

The boundary density is

$$e_m + f \cdot n = \frac{1}{2} \left[ (w_t - \partial_n w)^2 + |\nabla_{\tan} w|^2 + m^2 w^2 \right] \ge 0.$$

The inequality $|w_t F| \le (w_t^2 + F^2)/2$ therefore yields $E_m' \le E_m + \|F\|_2^2/2$, with the norm on the current shrinking ball. Multiplying by $e^{-t}$ and integrating from zero proves the stated estimate.

If the source and both data vanish, the estimate gives $E_m(t) = 0$. Continuity implies $w_t = 0$ on each open shrinking ball. For an interior cone point $(t, x)$, the segment $(s, x)$, $0 \le s \le t$, stays in the cone. Hence $w(t,x) = w(0,x) + \int_0^t w_t(s,x)\mathrm{d}s = 0$. This retains the initial displacement, which energy alone would miss for a massless constant. Continuity gives zero on the lateral boundary and vertex. Every integral used a compact cone slice; no all-space integrability was needed.

**Exercise 5.2 (Finite speed for a semilinear equation, conditional on solutions).** Let $V \in C^2(\mathbb{R})$ and suppose two real $C^2$ fields $\phi_1, \phi_2$ solve $\Box\phi + V'(\phi) = 0$ on a neighbourhood of the same closed cone. Assume their displacements and velocities agree on $B_R(a)$ at time zero.

(a) Prove that the fields agree on the closed cone. Use the mass-one energy of their difference, allowing $V''$ to have either sign. Give an explicit finite coefficient in the energy inequality in terms of a bound for $|V''|$ on the field values occurring in the cone.

(b) Suppose instead that the fields are defined on an open slab $I \times \mathbb{R}^d$, with $0 \in I$, and their two initial data agree outside a closed set $K$. Prove agreement at every $(t, x)$ with $\operatorname{dist}(x, K) > |t|$. What extra condition makes the zero field an available comparison solution?

**Solution.** Put $w = \phi_1 - \phi_2$. The ordinary fundamental theorem gives

$$b(t,x) = \int_0^1 V''(\phi_2 + s(\phi_1 - \phi_2))\mathrm{d}s, \qquad \Box w + bw = 0.$$

The fields are bounded on the compact cone. Their values and every segment between them lie in some compact interval $[-M, M]$. Thus $L = \max_{|z|\le M} |V''(z)|$ is finite and $|b| \le L$. No positivity of $b$ is required. Write the equation as $(\Box + 1)w = (1 - b)w$. The exact balance from the first problem gives

$$E_1'(t) \le \int_{B_{R-t}} |1 - b|\,|w_t w|\mathrm{d}^d x \le \frac{1+L}{2} \int_{B_{R-t}} (w_t^2 + w^2)\mathrm{d}^d x \le (1+L)E_1(t).$$

Matching smooth initial displacement also gives matching initial spatial gradients on the open ball, so $E_1(0) = 0$. The integrating factor $e^{-(1+L)t}$ yields $E_1 = 0$. Its mass term gives $w = 0$ on the interior directly; continuity includes the cone boundary and vertex.

For a later event outside the asserted influence set, the two data coincide on a neighbourhood of $\overline{B_t(x)}$. Apply the preceding argument to the compact cone of height $t$ with this base and the event as vertex. It lies in the open slab. Thus the fields agree at the event. Reverse time for earlier events. No bound uniform over all of space is needed: a separate compact cone supplies a finite $L$ for each event. The constant field zero solves the equation exactly when $V'(0) = 0$. Under that condition the result also gives finite propagation of the support of any existing solution with compactly supported data. None of these arguments constructs such a solution.

**Exercise 5.3 (Huygens' principle and the dimension).** Consider the massless wave with zero displacement and real $g \in C_c^\infty(\mathbb{R}^d)$ as initial velocity. The following classical representation formulas may be used as given facts for $t > 0$:

$$u_1(t,x) = \frac{1}{2}\int_{x-t}^{x+t} g(y)\,\mathrm{d}y, \qquad u_2(t,x) = \frac{1}{2\pi}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^2y,$$

$$u_3(t,x) = \frac{1}{4\pi t}\int_{|y-x|=t} g(y)\,\mathrm{d}S_y.$$

For the higher-dimensional classification one may use these general forms. Let $\alpha_d$ be the volume of the unit ball, $\fint$ the integral divided by the region's volume or area, and $D_t = t^{-1}\partial_t$. For odd $d \geq 3$ and even $d \geq 2$, respectively,

$$u_d = \frac{1}{1\cdot 3\cdots(d-2)} D_t^{(d-3)/2}\left[t^{d-2}\fint_{|y-x|=t} g(y)\,\mathrm{d}S_y\right],$$

$$u_d = \frac{1}{2\cdot 4\cdots d} D_t^{(d-2)/2}\left[\frac{1}{\alpha_d}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^dy\right].$$

These facts hold for the smooth compact data specified here; see Evans, *Partial Differential Equations*, 2nd ed., §2.4.1, formulas (31), (38), Theorems 2–3, pp. 73–77. Nonzero displacement contributes the time derivative of the corresponding velocity formula with $g$ replaced by that displacement.

(a) For $d = 1, 2, 3$, let $g \geq 0$, $g \not\equiv 0$, with support inside $B_1(0)$. Decide whether $u_d(t,0)$ vanishes for $t > 1$. Explain the difference between finite speed and dependence only on data arbitrarily near the base sphere.

(b) Use the general formulas to classify all positive spatial dimensions. For even $d$, test data supported strictly inside the base sphere so that every time derivative acts on a smooth kernel. Explain why a statement that all even-dimensional kernels are positive would be too strong.

**Solution.** In one dimension the interval $[-t,t]$ contains the support for $t > 1$. Thus $u_1(t,0) = \frac{1}{2}\int g > 0$ even though $g$ is zero near its two endpoints. In two dimensions the kernel is strictly positive on the fixed support, so $u_2(t,0) > 0$. In three dimensions the sphere of radius $t > 1$ misses the support, hence $u_3(t,0) = 0$. All three respect finite speed: their values use only data in the closed base ball. Sphere dependence is the stronger statement that data vanishing near the base sphere have zero effect at the event, even when they are nonzero in its interior.

For odd $d \geq 3$, the supplied formula uses a finite number of time derivatives of spherical averages. If both data vanish on a neighbourhood of the sphere at the specified time, these averages vanish on a time neighbourhood there, and all their derivatives vanish. This proves the sphere property, also including the given displacement term.

For even $d = 2\ell + 2$, put the nonnegative nonzero data inside $B_1(0)$ and fix $t > 1$. The support stays strictly away from the moving boundary for nearby times, so differentiation under the fixed-support integral is valid. Induction gives, with the empty product equal to one,

$$D_t^\ell (t^2 - r^2)^{-1/2} = (-1)^\ell (1\cdot 3\cdots(2\ell-1))(t^2-r^2)^{-\ell-1/2}.$$

The coefficient is nonzero and the remaining kernel is strictly positive on the support. Its integral against $g$ cannot vanish. Thus every even dimension has an interior-tail counterexample to the sphere property. The sign alternates, becoming negative already for $d = 4$; positivity of the two-dimensional kernel must not be asserted in all even dimensions.